

```
C:\Documents and Settings\user>ping 10.0.0.10
```

```
Pinging 10.0.0.10 with 32 bytes of data:
```

```
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123  
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
```

```
Ping statistics for 10.0.0.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

```
C:\Documents and Settings\user>ping 15.0.0.10
```

```
Pinging 15.0.0.10 with 32 bytes of data:
```

```
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
```

```
Ping statistics for 15.0.0.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

```
C:\Documents and Settings\user>
```

```
R2#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 30.0.0.2  
S    15.0.0.0/8 [1/0] via 30.0.0.2  
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C     20.0.0.0/8 is directly connected, Ethernet0/0  
L     20.0.0.2/32 is directly connected, Ethernet0/0  
S    21.0.0.0/8 [1/0] via 30.0.0.2  
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C     30.0.0.0/8 is directly connected, Ethernet0/1  
L     30.0.0.1/32 is directly connected, Ethernet0/1  
S    162.14.0.0/16 [1/0] via 30.0.0.2  
S    172.19.0.0/16 [1/0] via 30.0.0.2  
S    192.168.168.0/24 [1/0] via 30.0.0.2
```

```
R6#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 192.168.168.1  
S    15.0.0.0/8 [1/0] via 192.168.168.1  
S    20.0.0.0/8 [1/0] via 192.168.168.1  
S    21.0.0.0/8 [1/0] via 192.168.168.1  
S    30.0.0.0/8 [1/0] via 192.168.168.1  
    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks  
C        162.14.0.0/16 is directly connected, Ethernet0/0  
L        162.14.0.1/32 is directly connected, Ethernet0/0  
S    172.19.0.0/16 [1/0] via 192.168.168.1  
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks  
C        192.168.168.0/24 is directly connected, Ethernet0/1  
L        192.168.168.2/32 is directly connected, Ethernet0/1  
R6#
```

```
C:\Documents and Settings\user>ping 10.0.0.10
```

```
Pinging 10.0.0.10 with 32 bytes of data:
```

```
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
```

```
Ping statistics for 10.0.0.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

```
C:\Documents and Settings\user>ping 162.14.0.10
```

```
Pinging 162.14.0.10 with 32 bytes of data:
```

```
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124  
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124  
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124  
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124
```

```
Ping statistics for 162.14.0.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

```
C:\Documents and Settings\user>_
```

```
C:\Documents and Settings\user>ping 15.0.0.10
```

```
Pinging 15.0.0.10 with 32 bytes of data:
```

```
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124  
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
```

```
Ping statistics for 15.0.0.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
```

```
C:\Documents and Settings\user>ping 162.14.0.10
```

```
Pinging 162.14.0.10 with 32 bytes of data:
```

```
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123  
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123  
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123  
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
```

```
Ping statistics for 162.14.0.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

```
C:\Documents and Settings\user>
```

```
R5#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 21.0.0.1  
      [1/0] via 20.0.0.1  
S    15.0.0.0/8 [1/0] via 21.0.0.1  
      [1/0] via 20.0.0.1  
S    20.0.0.0/8 [1/0] via 21.0.0.1  
21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C    21.0.0.0/8 is directly connected, Ethernet0/0  
L    21.0.0.2/32 is directly connected, Ethernet0/0  
S    30.0.0.0/8 [1/0] via 21.0.0.1  
S    162.14.0.0/16 [1/0] via 192.168.168.2  
S    172.19.0.0/16 [1/0] via 21.0.0.1  
      [1/0] via 20.0.0.1  
192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks  
C    192.168.168.0/24 is directly connected, Ethernet0/1  
L    192.168.168.1/32 is directly connected, Ethernet0/1
```

```
R5#
```

```
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
     21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      21.0.0.0/8 is directly connected, Ethernet0/2
L      21.0.0.1/32 is directly connected, Ethernet0/2
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/8 is directly connected, Ethernet0/0
L      30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
     172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.19.0.0/16 is directly connected, Ethernet0/1
L      172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
```

```
R3#
```

```
R4#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 172.19.0.1  
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C    15.0.0.0/8 is directly connected, Ethernet0/0  
L    15.0.0.1/32 is directly connected, Ethernet0/0  
S    20.0.0.0/8 [1/0] via 172.19.0.1  
S    21.0.0.0/8 [1/0] via 172.19.0.1  
S    30.0.0.0/8 [1/0] via 172.19.0.1  
S    162.14.0.0/16 [1/0] via 172.19.0.1  
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks  
C    172.19.0.0/16 is directly connected, Ethernet0/1  
L    172.19.0.2/32 is directly connected, Ethernet0/1  
S    192.168.168.0/24 [1/0] via 172.19.0.1
```

```
R4#
```

```

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       21.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       162.14.0.0/16 [1/0] via 20.0.0.2
S       172.19.0.0/16 [1/0] via 20.0.0.2
S       192.168.168.0/24 [1/0] via 20.0.0.2
R1#

```